MILITARY SPECIFICATION

TRANSMITTER, PRESSURE, FUEL, 0–600 PSI

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of fuel pressure transmitter designated MS28005–7.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP–B–636—Box, Fiberboard.

MILITARY

MIL–P–116—Preservation, Methods of.
MIL–L–5020—Liquid, Compass, Aircraft.
MIL–E–5272—Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL–T–5350—Transmitters, Synchro, Aircraft, General Specification for.
MIL–I–6181—Interference Control Requirements, Aircraft Equipment.
MIL–I–6301—Indicator, Pressure, Fuel, 0–600 PSI.
MIL–P–7936—Parts and Equipment, Aeronautical, Preparation for Delivery.

STANDARDS

MILITARY

MIL–STD–129—Marking for Shipment and Storage.
MIL–STD–130—Identification Marking of U.S. Military Property.
MS28005—Transmitters, Pressure, Synchro, Aircraft.
MS33656—Fitting End, Standard Dimensions for Flared Tube Connection and Gasket Seal.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The transmitters furnished under this specification shall be a product which has been subjected to and which has passed the qualification tests specified herein, and is listed on or approved for listing on the applicable Qualified Products List.

3.2 General specification. The requirements of Specification MIL–T–5350 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction. The fuel pressure transmitter shall be designed for use with a fuel pressure indicator conforming to Specification MIL–I–6301.

3.3.1 *Hermetically sealed transmitter case.* The case shall provide a hermetically sealed (see 6.3) enclosure for the entire internal mechanism of the transmitter. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which may be affected by the action of any atmosphere to which the transmitter may be subjected.

3.3.1.1 *Case pressure release.* A metal blowout device to discharge excessive case pressure shall be provided in the pressure transmitter. The blowout device shall discharge to a $\frac{5}{16}$ –24

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fitting conforming to Standard MS33656. The device shall operate to release the internal case pressure through an orifice, approximately 0.045 inch in diameter, whenever the internal case pressure reaches 90±30 pounds per square inch (psi). The device shall not affect hermetic sealing. The design of the blowout device shall be submitted to the procuring activity for prior approval.

3.3.1.2 *Material.* The hermetically sealed case shall be constructed of a suitable material applicable to hermetic sealing, and the material shall be uniform in texture with a smooth external surface.

3.3.1.3 *Joints.* All joints subjected to pressure shall be joined with silver solder.

3.3.1.4 *Case proof pressure.* The transmitter case shall be so constructed that it shall be able to sustain an internal pressure of 135 psi without leakage from or distortion of the case.

3.3.2 *Operation.* The transmitter shall be capable of operating in either a horizontal or vertical position.

3.4 Dimensions. The outline dimensions of the transmitter shall conform to Part No. MS28005–7.

3.5 Weight. The weight of the transmitter shall not exceed 1.5 pounds.

3.6 Overpressure stop. An overpressure stop shall be provided and shall be designed to restrain the prime mover. The stop shall allow the pointer of the indicator to travel past full scale to a point approximately midway between the last and the zero graduation.

3.7 Zero reset. The transmitter shall be provided with means to adjust or correct the calibration without breaking a solder-sealed area. The means of adjustment shall be accessible from the exterior of the case and shall provide a minimum total of 15 degrees of synchro adjustment. The location of the adjustment shall be optional.

3.7.1 *Compressed scale adjustment.* In the event that a separate adjustment is necessary, the transmitter shall be provided with a means to adjust or correct the calibration of the compressed portion of the scale. The means of adjustment shall be accessible from the exterior of the case and be accomplished without breaking a solder-sealed area. The location of the compressed scale adjustment shall be optional.

3.8 The straight thread connections shall be marked "P" and "V" in durable white letters ⅛ inch in height, as indicated by Standard MS28005.

3.9 Identification of product. Equipment assemblies, and parts shall be marked for identification in accordance with Standard MIL–STD–130 and shall include:

TRANSMITTER, PRESSURE, FUEL, 0–600 PSI MS28005–7
26V, single phase, 400 cycles

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of transmitters shall be classified as:

(a) Qualification tests ..... (4. 3)
(b) Quality conformance tests ..... (4. 4)

4.3 Qualification tests.

4.3.1 *Sampling instructions.* The qualification test samples shall consist of three transmitters. Samples shall be identified with the manufacturer's own part number and any additional information requested by the letter of authorization (see 6.4).

4.3.2 *Tests.* The qualification tests shall consist of all the examinations and tests specified under 4.6.

4.4 Quality conformance tests. The quality conformance tests shall consist of individual tests and sampling plans and tests (see 4.4.1 and 4.4.2).

4.4.1 *Individual tests.* Each transmitter shall be subjected to:

(a) Detail examination ..... (4. 6. 1)
(b) Scale error at room temperature ..... (4. 6. 2)
(c) Friction ..... (4. 6. 3)
(d) Position error ..... (4. 6. 4)
(e) Dielectric strength ..... (4. 6. 5)
(f) Leak ..... (4. 6. 6)

4.4.2 *Sampling plans and tests.*

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4.4.2.1 *Sampling plan A tests.* Five transmitters selected at random from each 500 or less on the contract or order shall be subjected to:

(a) Individual tests ..... (4. 4. 1)
(b) Low temperature ..... (4. 6. 7)
(c) High temperature ..... (4. 6. 8)
(d) Vibration error ..... (4. 6. 9)
(e) Electrical characteristics ... (4. 6. 10)
(f) Differential pressure error .. (4. 6. 11)

4.4.2.2 *Sampling plan B tests.* Unless otherwise specified (see 6.2), five transmitters selected at random from the first 25 items on the contract or order shall be subjected to:

(a) Sampling plan A tests ..... (4. 4. 2. 1)
(b) Torque ..... (4. 6. 12)
(c) Extreme high-temperature exposure ..... (4. 6. 13)
(d) Seasoning ..... (4. 6. 14)
(e) Endurance ..... (4. 6. 15)
(f) Radio noise suppression ... (4. 6. 16)
(g) Overpressure ..... (4. 6. 17)
(h) Salt spray ..... (4. 6. 18)
(i) Case pressure release ..... (4. 6. 19)
(j) Case proof pressure ..... (4. 6. 20)

4.5 Test conditions. Test conditions shall be in accordance with Specification MIL–T–5350 and as specified herein. Unless otherwise specified, the transmitter shall be tested with its longitudinal axis in the horizontal plane. Compass liquid in accordance with Specification MIL–L–5020, or any suitable liquid approved by the procuring activity, shall be used for testing purposes.

4.6 Test methods.

4.6.1 *Detail examination.* The transmitter shall be inspected to determine compliance with regard to workmanship, materials, and marking.

4.6.2 *Scale error at room temperature.* The transmitter shall be tested for scale error at room temperature at the test points specified in table I with both increasing and decreasing rates of flow. The flow with an increasing rate shall be measured by bringing the flow up to, but not exceeding, the specified rate. With a decreasing rate, the flow shall be brought down to, but not below, the specified rate. The transmitter shall be vibrated at a frequency of 3,600 cycles per minute (cpm) with an amplitude not to exceed 0.002 inch. The scale errors shall not exceed the tolerances specified in table I.

TABLE I. *Scale error tolerances*

Test points	Scale error (room temperature)	Vibration and high temperature	Seasoning, endurance, overpressure, altitude low temperature	Extreme high temperature
Psi	±Tolerance (psi)	±Tolerance (psi)	±Tolerance (psi)	±Tolerance (psi)
0	1. 0	3	3	5. 0
25	1. 5	3	3	5. 5
50	2. 0	3	4	6. 0
75	2. 5	5	6	6. 5
100	3. 0	6	6	7. 0
200	15. 0	20	25	35
300	15. 0	20	25	35
400	15. 0	25	30	35
500	20. 0	30	40	40
600	20. 0	30	40	40

4.6.3 *Friction.* The difference in reading prior to application of vibration specified in 4.6.2 is friction error, and shall not exceed the tolerance specified in table II.

TABLE II. *Pointer error and friction error tolerances*

Test	± Tolerance (psi)	
	Range 0–100	Over 100
Friction:		
Room temperature	2. 0	11. 0
High temperature	2. 0	11. 0
Low temperature	4. 0	20. 0
Position error at 300 psi	------	10. 0
Vibration:		
Pointer oscillation at 300 psi	------	15. 0

4.6.4 *Position error.* The position error test may be conducted in conjunction with the test specified in 4.6.2. The transmitter shall be energized to obtain a reading of approximately 300 psi. The change in indication produced by rotating the transmitter from the normal operating position to a position 90 degrees clockwise and counterclockwise (longitudinal axis vertical) shall be noted. The maximum change in indication shall not exceed the tolerance specified for position error in table II.

4.6.5 *Dielectric strength.* A potential of 200 volts at a commercial frequency shall be applied between any pin of the electrical receptacle and

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the transmitter case for a minimum period of 5 seconds. There shall be no breakdown of the insulation.

4.6.6 *Leak.* Sensitive pressure-sensing elements, with a range appropriate for the pressure to be applied, shall be teed into the vent and pressure lines. A negative pressure of 14 psi shall be applied to the vent connection, and a pressure of 400 psi shall be applied to the pressure connection. The pressure lines shall be sealed at a point within approximately 2 inches of each connection. During a period of 1 minute, the pressure-sensing elements shall show no change in indication.

4.6.7 *Low temperature.* The transmitter shall be placed in a chamber and subjected to a temperature of $-54° \pm 2°$ C for a period of 4 hours. At the end of this period and while still at this temperature, the transmitter shall be subjected to a friction and scale error test. The scale errors shall not exceed the tolerance specified in table I and the friction errors shall not exceed the tolerance specified in table II. After the transmitter has returned to room temperature, it shall again be subjected to the scale and friction error tests. The errors shall not exceed the applicable tolerances of tables I and II.

4.6.8 *High temperature.* The transmitter shall be placed in a chamber and subjected to a temperature of $70° \pm 2°$ C for a period of 6 hours. The transmitter shall be electrically connected to an appropriate indicator with power applied during this test. At the end of this period and with the transmitter still at this temperature, the transmitter shall be subjected to a friction and scale error test. The scale errors shall not exceed the tolerance specified in table I and the friction errors shall not exceed the tolerance specified in table II. After the transmitter has returned to room temperature, it shall again be subjected to the scale and friction error tests. The errors shall not exceed the applicable tolerances of tables I and II.

4.6.9 *Vibration.* The transmitter shall be subjected to vibration for a period of 3 hours on a suitably designed vibration stand at an amplitude of 0.018 to 0.020 inch at a frequency range from 300 to 3,000 cpm. During this period, a pressure shall be applied to the transmitter sufficient to produce approximately half-scale deflection. The amplitude of pointer oscillation of a master indicator shall not exceed the tolerance specified in table II. The frequency shall be varied from 300 to 3,000 cpm, and observation shall be made for any resonant points. If the transmitter does not indicate any resonance within the frequency range, the frequency shall be set at any point between 2,000 and 3,000 cpm. If resonance is noted, the transmitter shall be vibrated at that point for the duration of the test. Following the vibration period, the transmitter shall be tested for scale errors as specified in 4.6.2, on increasing readings only. Errors shall not exceed the tolerances specified in table I.

4.6.10 *Electrical characteristics.* The transmitter shall be tested for compliance with the requirements specified in Specification MIL–T–5350 for primary current, power input, secondary voltage, and for operating either one or two indicators without a change in indication of more than 1 degree of arc.

4.6.11 *Differential pressure error.* The transmitter shall be tested for scale errors with a vacuum of 10 psi applied to the vent connection of the transmitter. The test shall be accomplished by subjecting the transmitter to the pressures specified to produce the indicator readings shown in table III. Errors shall not exceed the tolerances specified in table III.

TABLE III. *Differential pressure error*

Vent connection vacuum (psi) (gage)	Pressure connection pressure (psi) (gage)	Gage indication (psi)	Tolerance (psi)
10.0	40. 0	50	5. 0
10.0	90. 0	100	5. 0
10.0	190. 0	200	15. 0
10.0	290. 0	300	20. 0
10.0	390. 0	400	25. 0
10.0	490. 0	500	25. 0

4.6.12 *Torque.* The three secondary leads of the transmitter synchro shall be connected to the corresponding leads of a suitable indicator synchro. Normal excitation of 26 volts, 400 cps shall be supplied to the primaries of both synchros. The pointer of the indicator synchro shall be rotated through an angle from its corresponding synchronous position and the

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torque tending to return the pointer of the indicator synchro to its synchronous position shall be measured. The torque is directly proportional to the angular displacement within displacement up to 20 degrees and shall be no less than 2,000 milligram-millimeters per degree of displacement.

4.6.13 *Extreme high-temperature exposure.* The transmitter shall be subjected to a temperature of 110° ±3° C for a period of 48 hours. At the end of this period and while at this temperature, the transmitter shall be subjected to the scale error and friction tests (4.6.2 and 4.6.3). The scale error shall not exceed the tolerance specified for extreme high temperature in table I and the friction error shall not exceed the tolerance specified in table II. After the transmitter has returned to room temperature, it shall again be subjected to the scale and friction error tests. The errors shall not exceed the applicable tolerances of tables I and II.

4.6.14 *Seasoning.* The transmitter shall be operated to indicate the equivalent of zero indication to 80 ±5 percent full scale and back to zero indication for 10,000 times at a rate of 30±10 cpm. The transmitter shall be tested not less than 1 hour after this test for scale errors. Errors shall not exceed the tolerances specified in table I. The transmitter shall be examined to determine if undue wear has occurred.

4.6.15 *Endurance.* The transmitter shall be subjected to 100,000 applications of pressure to produce an indication from zero to 300 psi and back to zero at a rate of 30 ±10 cpm. The transmitter shall be tested not less than 1 hour after this test for scale errors at room temperature (4.6.2) and errors shall not exceed the tolerances specified in table I. Transmitters subjected to this test shall be examined to determine whether undue wear has occurred.

4.6.16 *Radio noise suppression.* Radio noise suppression shall be in accordance with Specification MIL–I–6181, except that a frequency range of from 0.15 to 150 megacycles shall apply for the radiated radio noise test.

4.6.17 *Overpressure.* The transmitter shall be subjected to a pressure of 800 psi for a period of 10 minutes. Not less than 1 hour following the application of the overpressure, the transmitter shall be tested for scale errors on increasing pressures only. Errors shall not exceed the tolerances specified in table I.

4.6.18 *Salt spray.* The transmitter shall be subjected to the salt spray test, in accordance with Procedure I of Specification MIL–E–5272, for a period of 50 hours. The transmitter shall be examined, and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation.

4.6.19 *Case pressure release.* The pressure blowout device shall be tested either under simulated increasing case pressure or by actual injection of pressure into the transmitter case. The device shall operate at a pressure of 90 ±30 psi.

4.6.20 *Case proof pressure.* Following the case pressure release test (4.6.19), the case pressure release fitting shall be closed off and the case pressurized to 135 psi. Any leaks in, or distortion of, the case shall be cause for rejection.

4.7 Preservation, packaging, packing, and marking. Preparation for delivery shall be examined for conformance to section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Each transmitter shall be preserved and packaged in accordance with Method IA of Specification MIL–P–116 for all levels of packing.

5.2 Packing. Transmitters shall be prepared for shipment in accordance with level A, B, or C, as specified (see 6.2), of Specification MIL–P–7936. Containers shall conform to Specification PPP–B–636.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with Standard MIL–STD–129. Each container shall be marked with the following information:

TRANSMITTER, PRESSURE, FUEL, SYNCHRO, 0–600 PSI
320-DEGREE MOVEMENT, MS28005–7, 26V,
SINGLE PHASE, 400 CYCLES
Specification MIL–T–6302B

6. NOTES

6.1 Intended use. The fuel pressure transmitter is intended for use to transmit the indicated pressure in the fuel system of aircraft engines to an applicable indicator.

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6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.

(b) Whether or not sampling plan B is to be omitted (see 4.4.2.2).

(c) Applicable level of packing (see 5.2).

6.3 Definition.

6.3.1 *Hermetic seal.* A hermetic seal is defined as a perfectly closed and airtight seal made between vitric, metallic, or between vitric and metallic materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Middletown Air Materiel Area, Attn: MANEE, Olmsted Air Force Base, Pennsylvania, and information pertaining to qualification of products may be obtained from that activity.

6.5 Marginal indicia. The margins of this specification are marked to indicate where changes, deletions, or additions to the previous issue have been made. This is done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Figures are not so marked. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

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Project No. 6620–0057.

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SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL–T–6302B Transmitter, Pressure, Fuel, 0–600 PSI

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS OF ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *If "yes" in what way?*

4. REMARKS

(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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